Temperature and Humidity Monitoring System

Arduino Uno R3 • DHT11 sensor • Serial Monitor output (no LCD)

```
/*
  Project: Temperature and Humidity Monitoring System
  Board: Arduino Uno R3
  Sensor: DHT11 (change to DHT22 if you use a DHT22)
  Output: Arduino IDE Serial Monitor (no LCD required)

  Typical DHT11 wiring:
    DHT VCC  -> Arduino 5V
    DHT GND  -> Arduino GND
    DHT DATA -> Arduino digital pin 2

  If you have a bare 4-pin DHT sensor, use a 10 kOhm pull-up resistor
  between VCC and DATA. Many sensor modules already include one.

  Arduino IDE library required:
    Install "DHT sensor library" by Adafruit.
    If prompted, also install "Adafruit Unified Sensor".
*/

#include <DHT.h>

#define DHTPIN 2       // DHT sensor data pin connected to digital pin 2
#define DHTTYPE DHT11  // Change to DHT22 if you have a DHT22 sensor

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();

  Serial.println("Temperature and Humidity Monitoring System");
  Serial.println("------------------------------------------");
  delay(2000);
}

void loop() {
  // DHT sensors need time between readings; 2 seconds is a safe interval.
  float humidity = dht.readHumidity();
  float temperatureC = dht.readTemperature();

  if (isnan(humidity) || isnan(temperatureC)) {
    Serial.println("Failed to read from DHT sensor. Check wiring and sensor type.");
    delay(2000);
    return;
  }

  Serial.print("Temperature: ");
  Serial.print(temperatureC, 1);
  Serial.print(" C  |  Humidity: ");
  Serial.print(humidity, 1);
  Serial.println(" %");

  delay(2000);
}
```

Setup notes: Install Adafruit's “DHT sensor library” in Arduino IDE (and “Adafruit Unified Sensor” if prompted). Connect sensor DATA to digital pin 2, VCC to 5V, and GND to GND. Open Serial Monitor at 9600 baud. For a DHT22 sensor, change `DHTTYPE DHT11` to `DHTTYPE DHT22`.